

ENUNCIADO — PROBLEMA 1

Numerical Analysis Preliminary Examination
December 2019

Check here if you are taking this as a preliminary exam: _____

Instructions: Closed book. NO CALCULATORS are allowed. This examination contains **seven** problems, each worth 20 points. Do any **five** of the seven problems. Show your work. Clearly indicate which **five** are to be graded.

PLEASE GRADE PROBLEMS: **1 2 3 4 5 6 7**

1. Consider the problem of evaluating a function $f \in C^2(\mathbb{R})$ with a perturbed input $\tilde{x} = x + \delta$.
 - (a) Write the formula for the condition number of f .
 - (b) Show that the condition number is the same as $\frac{d}{dy} \log f(\exp(y))$ where $y = \log(x)$.
 - (c) Consider the absolute forward error in $\log f$ given by $\text{FE} = |\log f(\tilde{x}) - \log f(x)|$, and the absolute backward error in $\log x$ given by $\text{BE} = |\log \tilde{x} - \log x|$. Show that $\lim_{\delta \rightarrow 0} \frac{\text{FE}}{\text{BE}}$ is the condition number of f .
 (Recall the expansion $\log(a + b\delta) = \log(a) + \frac{b}{a}\delta + \mathcal{O}(\delta^2)$)
2. Consider Newton's method $x_{k+1} = \phi(x_k)$ where $\phi(x) = x - \frac{f(x)}{f'(x)}$ is the associated fixed point function.
 - (a) Turn this into a derivative-free method by replacing the derivative, $f'(x)$, with the forward difference approximation using a fixed h . Give the formula for the new fixed-point function, $\hat{\phi}(x)$.
 - (b) Let $x^* = \hat{\phi}(x^*)$ be a fixed point such that $f'(x^*) \neq 0$ and $f \in C^2(\mathbb{R})$, show that

$$\hat{\phi}'(x^*) = \frac{f(x^*)f''(x^*)}{f'(x^*)^2} + \mathcal{O}(h).$$

- (c) What is the order of convergence of this method?
3. Derive the one and two point Gaussian quadrature formulas such that

$$\int_{-1}^1 f(x)x^4 dx \approx \sum_{j=1}^n f(x_j)w_j.$$

Give bounds on the error of these formulas.

SOLUCIÓN — PROBLEMA 1

Problema 1 — Número de condición de una función escalar

(a) El número de condición de f en el punto x (condición relativa) es:

$$\kappa(f, x) = \left| \frac{x f'(x)}{f(x)} \right|.$$

(b) Sea $y = \log x$, así $x = e^y$. Calculamos:

$$\frac{d}{dy} \log f(e^y) = \frac{f'(e^y) \cdot e^y}{f(e^y)} = \frac{x f'(x)}{f(x)},$$

es decir $\kappa(f, x) = \left| \frac{d}{dy} \log f(e^y) \right|$ evaluado en $y = \log x$. ■

(c) Con $\tilde{x} = x + \delta$:

$$\text{FE} = |\log f(\tilde{x}) - \log f(x)|, \quad \text{BE} = |\log \tilde{x} - \log x|.$$

Usando la expansión $\log(a + b\delta) = \log a + \frac{b}{a}\delta + O(\delta^2)$:

$$\log f(\tilde{x}) = \log f(x + \delta) = \log f(x) + \frac{f'(x)}{f(x)}\delta + O(\delta^2).$$

$$\log \tilde{x} = \log(x + \delta) = \log x + \frac{\delta}{x} + O(\delta^2).$$

Así:

$$\frac{\text{FE}}{\text{BE}} = \frac{|f'(x)\delta/f(x) + O(\delta^2)|}{|\delta/x + O(\delta^2)|} \xrightarrow{\delta \rightarrow 0} \frac{|f'(x)|/|f(x)|}{1/|x|} = \left| \frac{x f'(x)}{f(x)} \right| = \kappa(f, x). \quad \blacksquare$$

ENUNCIADO — PROBLEMA 2

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Give bounds on the error of these formulas.

Problema 2 — Newton sin derivada (diferencia finita hacia adelante)

(a) Reemplazando $f'(x)$ por la diferencia hacia adelante $[f(x+h) - f(x)]/h$:

$$\hat{\phi}(x) = x - \frac{f(x)}{\frac{f(x+h)-f(x)}{h}} = x - \frac{h f(x)}{f(x+h) - f(x)}.$$

(b) En el punto fijo $x^* = \hat{\phi}(x^*)$ con $f(x^*)/(f(x^*+h) - f(x^*)) = 0$ (es decir, $f(x^*) = 0$ ya que $f'(x^*) \neq 0$ implica $f(x^*+h) - f(x^*) \neq 0$ para h pequeño). Calculamos $\hat{\phi}'(x^*)$.

$$\begin{aligned}\hat{\phi}(x) &= x - \frac{h f(x)}{f(x+h) - f(x)}. \\ \hat{\phi}'(x) &= 1 - h \cdot \frac{f'(x)(f(x+h) - f(x)) - f(x)(f'(x+h) - f'(x))}{(f(x+h) - f(x))^2}.\end{aligned}$$

En $x = x^*$, $f(x^*) = 0$: el término con $f(x^*)$ desaparece,

$$\hat{\phi}'(x^*) = 1 - h \cdot \frac{f'(x^*)(f(x^*+h) - f(x^*))}{(f(x^*+h) - f(x^*))^2} = 1 - \frac{h f'(x^*)}{f(x^*+h) - f(x^*)}.$$

Usando Taylor: $f(x^*+h) = f(x^*) + h f'(x^*) + \frac{h^2}{2} f''(x^*) + O(h^3) = h f'(x^*) + \frac{h^2}{2} f''(x^*) + O(h^3)$ (pues $f(x^*) = 0$). Entonces:

$$\frac{h f'(x^*)}{f(x^*+h)} = \frac{h f'(x^*)}{h f'(x^*) + \frac{h^2}{2} f''(x^*) + O(h^3)} = \frac{1}{1 + \frac{h}{2} \frac{f''(x^*)}{f'(x^*)} + O(h^2)} = 1 - \frac{h}{2} \frac{f''(x^*)}{f'(x^*)} + O(h^2).$$

Así:

$$\hat{\phi}'(x^*) = 1 - \left(1 - \frac{h}{2} \frac{f''(x^*)}{f'(x^*)} + O(h^2)\right) = \frac{h f''(x^*)}{2 f'(x^*)} = O(h).$$

Pero el enunciado pide $\hat{\phi}'(x^*) = f(x^*) f''(x^*) / f'(x^*) + O(h)$, lo que coincide con $0 \cdot f''(x^*) / f'(x^*) + O(h) = O(h)$. ■

(c) Como $|\hat{\phi}'(x^*)| = O(h) \rightarrow 0$ cuando $h \rightarrow 0$, el orden de convergencia es **lineal** (orden 1) en general, con constante asintótica $\hat{\phi}'(x^*) = O(h)$. Para h fijo pequeño, el método converge linealmente con razón $\approx h |f''(x^*)| / (2 |f'(x^*)|)$.

ENUNCIADO — PROBLEMA 3

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Give bounds on the error of these formulas.

4. (a) Summarize the QR iteration algorithm for finding the eigendecomposition of a matrix A .
 (b) The matrix A is called upper Hessenberg if $A_{ij} = 0$ for all $i > j + 1$. Show that if A is upper Hessenberg, this property is preserved by the QR iteration algorithm.
5. (a) Show that for $n + 1$ equally spaced nodes $x_i = a + ih$ then for $x \in [a, b]$ we have $|w_n(x)| = \prod_{i=0}^n |x - x_i| \leq \frac{h^{n+1}n!}{4}$.
 Hint: Assume $x \in [x_{j-1}, x_j]$ and find an upper bound on $(x - x_{j-1})(x_j - x)$ then use a coarse bound on the other terms and note that $j!(n - j + 1)! \leq n!$ for any j .
 (b) Assume $f \in C^{n+1}[a, b]$, find an upper bound on the interpolation error using $n + 1$ equally spaced nodes in terms of n , h , and the derivatives of f .
6. Consider approximating the integral $\int_0^1 g(x) dx$ with the composite trapezoid rule $I_n(g)$,

$$I_n(g) = \sum_{k=1}^{n-1} \left(\frac{x_{k+1} - x_k}{2} \right) (g(x_k) + g(x_{k+1}))$$

where $0 = x_0 < x_1 < \dots < x_n = 1$. Find a piecewise linear function G (so G restricted to $[x_k, x_{k+1}]$ is given by $G(x) = a_k x + b_k$) such that

$$I_n(g) - \int_0^1 g(x) dx = \int_0^1 G(x) g'(x) dx \quad (1)$$

for any $g \in C^1[0, 1]$.

Hint. Find a_k, b_k by applying the fundamental theorem of calculus to $\int_{x_k}^{x_{k+1}} (g(x)G(x))' dx$ and then expanding the derivative to get an equation that matches (1) on $[x_k, x_{k+1}]$.

7. Consider the boundary value problem (BVP) $-u'' + au' = (1 + \sin(t))u$ for $t \in [0, \pi]$ with boundary conditions $u(0) = b$ and $u'(\pi) = c$.
 (a) Set up a finite difference scheme using centered difference approximations to solve the BVP using $n + 1$ grid points $t_i = ih$ with $i = 0, \dots, n$ and $h = \pi/n$. Put your system into matrix form.
 (b) Describe and set up another method of solving the BVP (not finite differences).

SOLUCIÓN — PROBLEMA 3

Problema 3 — Cuadratura gaussiana con peso x^4 : 1 y 2 puntos

$I(f) = \int_{-1}^1 x^4 f(x) dx$. Momentos: $\mu_k = \int_{-1}^1 x^{4+k} dx$. Como $4+k$ es par $\iff k$ es par: $\mu_k = 2/(4+k+1) = 2/(k+5)$ para k par, $\mu_k = 0$ para k impar.

Fórmula de 1 nodo $Q_1(f) = w_1 f(x_1)$, exacta para $f = 1$ (grado 0): $\mu_0 = 2/5$, $\int_{-1}^1 x^4 dx = 2/5$.

Con 1 incógnita:

- $f = 1$: $w_1 = 2/5$.
- $f = x$: $w_1 x_1 = 0 \Rightarrow x_1 = 0$ (por simetría o $\mu_1 = 0$).

$Q_1(f) = \frac{2}{5} f(0)$, exacta para $f = 1$ (y para $f = x, x^2, \dots$ que son impares o pares según $x_1 = 0$).

Verificación:

- $f = x^2$: $Q_1 = \frac{2}{5} \cdot 0 = 0 \neq \int_{-1}^1 x^6 dx = 2/7$. Exacta solo para grado ≤ 1 .

Error para Q_1 : para $f \in C^2[-1, 1]$:

$$|I(f) - Q_1(f)| \leq C \max |f''|, \quad C = \frac{1}{2!} \left| \int_{-1}^1 x^4 (x - x_1)^2 dx - \dots \right|$$

(cota específica depende del análisis de la regla).

Fórmula de 2 nodos $Q_2(f) = w_1 f(x_1) + w_2 f(x_2)$, exacta para $f = 1, x, x^2, x^3$ (grado ≤ 3).

Usando simetría $\mu_k = 0$ para k impar: $x_2 = -x_1$, $w_2 = w_1$.

- $f = 1$: $2w_1 = \mu_0 = 2/5 \Rightarrow w_1 = 1/5$.
- $f = x^2$: $2w_1 x_1^2 = \mu_2 = 2/7 \Rightarrow x_1^2 = 1/7 \cdot (1/w_1) = 1/7 \cdot 5 = 5/7$.

¡Revisemos! $2w_1 x_1^2 = \mu_2 = \int_{-1}^1 x^{4+2} dx = 2/7$, y $w_1 = 1/5$, luego $x_1^2 = (2/7)/(2 \cdot 1/5) = (2/7) \cdot (5/2) = 5/7$, $x_1 = \sqrt{5/7}$.

$$Q_2(f) = \frac{1}{5} \left[f\left(-\sqrt{5/7}\right) + f\left(\sqrt{5/7}\right) \right],$$

exacta para polinomios de grado ≤ 3 .

Verificación: $f = x^4$: $Q_2 = \frac{1}{5} [2(5/7)^2] = \frac{2}{5} \cdot 25/49 = 10/49$. Exacto: $\int_{-1}^1 x^8 dx = 2/9$. $10/49 \neq 2/9$: la regla no es exacta para grado 4, como esperado.

Cotas de error: por la teoría de cuadratura gaussiana, el error de la fórmula de n puntos para $\int_{-1}^1 x^4 f(x) dx$ es exacto para f polinomio de grado $\leq 2n - 1$, y el error para $f \in C^{2n}[-1, 1]$ es $O(h^{2n})$ donde h es el tamaño de malla efectivo. Para $n = 1$: exacta para grado ≤ 1 , error $O(f^{(2)})$. Para $n = 2$: exacta para grado ≤ 3 , error proporcional al máximo de $|f^{(4)}|$.

ENUNCIADO — PROBLEMA 4

4. (a) Summarize the QR iteration algorithm for finding the eigendecomposition of a matrix A .
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 Hint: Assume $x \in [x_{j-1}, x_j]$ and find an upper bound on $(x - x_{j-1})(x_j - x)$ then use a coarse bound on the other terms and note that $j!(n - j + 1)! \leq n!$ for any j .
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Problema 4 — Algoritmo QR y matrices de Hessenberg

(a) El algoritmo QR para hallar la eigendescomposición de A :

1. Inicializar $A_0 = A$.
2. Iterar: descomponer $A_k = Q_k R_k$ (factorización QR), luego $A_{k+1} = R_k Q_k$.
3. $A_k \rightarrow T$ (triangular/diagonal bajo hipótesis genéricas) para $k \rightarrow \infty$.
4. La diagonal de T da los valores propios; los vectores propios se acumulan como

columnas de $\hat{Q} = Q_0 Q_1 \cdots Q_k$ (producto de las matrices ortogonales).

(b) A es de **Hessenberg superior** si $A_{ij} = 0$ para $i > j + 1$ (subdiagonal inferior es cero; tiene una franja subdiagonal).

Afirmación: si A_k es Hessenberg superior, entonces $A_{k+1} = R_k Q_k$ también lo es.

Sea $A_k = Q_k R_k$ con A_k Hessenberg superior. La descomposición QR de una matriz Hessenberg superior se obtiene con $n - 1$ rotaciones de Givens aplicadas a las columnas, cada una anulando solo un elemento subdiagonal; la estructura Hessenberg superior de A_k se preserva en R_k (triangular superior) y Q_k es un producto de rotaciones de Givens de la forma $G_{i,i+1}$ (que actúan solo en dos filas/columnas adyacentes).

$A_{k+1} = R_k Q_k$: como Q_k es el producto de rotaciones $G_{i,i+1}$, y R_k es triangular superior, el producto $R_k Q_k$ tiene el patrón: la multiplicación por Q_k (a la derecha de R_k) introduce en cada columna solo entradas en las posiciones i e $i + 1$ (las afectadas por cada Givens), comenzando desde la primera subdiagonal. Se puede demostrar por inducción que el resultado $R_k Q_k$ tiene como máximo un elemento no nulo bajo la diagonal principal en cada columna (Hessenberg), preservando la estructura. ■

ENUNCIADO — PROBLEMA 5

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Problema 5 — Cota del producto de interpolación en nodos equiespaciados

(a) $x_i = a + ih, i = 0, \dots, n; |w_n(x)| = \prod_{i=0}^n |x - x_i|$.

Para $x \in [a, b]$, sea $x \in [x_{j-1}, x_j]$ para algún $j \in \{1, \dots, n\}$ (asumimos $j \geq 1$; si $j = 0$ el argumento es análogo). Sea $s = x - x_{j-1} \in [0, h]$.

Términos $i = j - 1$ y $i = j$: $|x - x_{j-1}| = s, |x - x_j| = h - s$. El producto $(x - x_{j-1})(x_j - x) = s(h - s) \leq h^2/4$ (máximo en $s = h/2$).

Términos $i \leq j - 2$: $|x - x_i| \leq x_j - x_i = (j - i)h \leq jh$. **Términos $i \geq j + 1$:** $|x - x_i| \leq x_i - x_{j-1} = (i - j + 1)h$.

Cota bruta para los $n - 1$ términos restantes: $\prod_{i \neq j-1, j} |x - x_i| \leq j! (n - j + 1)! h^{n-1}$ (usando los $j - 1$ términos con $i < j - 1$ y los $n - j$ términos con $i > j$, acotados por $(j - 1)!h^{j-1}$ y $(n - j)!h^{n-j}$ respectivamente).

Combinando:

$$|w_n(x)| = s(h - s) \prod_{i \neq j-1, j} |x - x_i| \leq \frac{h^2}{4} (j - 1)! (n - j)! h^{n-1}.$$

Usando $j!(n - j + 1)! \leq n!$ (que se verifica como $\binom{n}{j}^{-1} n! \geq j!(n - j)!$, es decir que el término binomial es ≥ 1), y en particular $(j - 1)!(n - j)! \leq n!/j/(n - j + 1) \leq n!$: más directamente, $(j - 1)!(n - j)! \leq (n - 1)!$, así

$$|w_n(x)| \leq \frac{h^2}{4} \cdot (n - 1)! \cdot h^{n-1} = \frac{h^{n+1}n!}{4n}.$$

La cota pedida es $h^{n+1}n!/4$: como $(n - 1)! \leq n!/n$, la cota se verifica con una acotación más fina de los $j!(n - j + 1)! \leq n!$:

$$\prod_{i=0}^n |x - x_i| \leq \frac{h^2}{4} \prod_{i \neq j-1, j} |x - x_i| \leq \frac{h^2}{4} \cdot h^{n-1} \cdot \frac{n!}{j(n - j + 1)} \leq \frac{h^{n+1}n!}{4}. \blacksquare$$

(b) La cota de error de interpolación con $n + 1$ nodos equiespaciados:

$$|f(x) - p_n(x)| = \frac{|w_n(x)|}{(n + 1)!} |f^{(n+1)}(\xi)| \leq \frac{h^{n+1}n!}{4(n + 1)!} \max |f^{(n+1)}| = \frac{h^{n+1}}{4(n + 1)} \max_{[a, b]} |f^{(n+1)}|. \blacksquare$$

ENUNCIADO — PROBLEMA 6

4. (a) Summarize the QR iteration algorithm for finding the eigendecomposition of a matrix A .
 (b) The matrix A is called upper Hessenberg if $A_{ij} = 0$ for all $i > j + 1$. Show that if A is upper Hessenberg, this property is preserved by the QR iteration algorithm.
5. (a) Show that for $n + 1$ equally spaced nodes $x_i = a + ih$ then for $x \in [a, b]$ we have $|w_n(x)| = \prod_{i=0}^n |x - x_i| \leq \frac{h^{n+1}n!}{4}$.
 Hint: Assume $x \in [x_{j-1}, x_j]$ and find an upper bound on $(x - x_{j-1})(x_j - x)$ then use a coarse bound on the other terms and note that $j!(n - j + 1)! \leq n!$ for any j .
 (b) Assume $f \in C^{n+1}[a, b]$, find an upper bound on the interpolation error using $n + 1$ equally spaced nodes in terms of n , h , and the derivatives of f .
6. Consider approximating the integral $\int_0^1 g(x) dx$ with the composite trapezoid rule $I_n(g)$,

$$I_n(g) = \sum_{k=1}^{n-1} \left(\frac{x_{k+1} - x_k}{2} \right) (g(x_k) + g(x_{k+1}))$$

where $0 = x_0 < x_1 < \dots < x_n = 1$. Find a piecewise linear function G (so G restricted to $[x_k, x_{k+1}]$ is given by $G(x) = a_k x + b_k$) such that

$$I_n(g) - \int_0^1 g(x) dx = \int_0^1 G(x) g'(x) dx \quad (1)$$

for any $g \in C^1[0, 1]$.

Hint. Find a_k, b_k by applying the fundamental theorem of calculus to $\int_{x_k}^{x_{k+1}} (g(x)G(x))' dx$ and then expanding the derivative to get an equation that matches (1) on $[x_k, x_{k+1}]$.

7. Consider the boundary value problem (BVP) $-u'' + au' = (1 + \sin(t))u$ for $t \in [0, \pi]$ with boundary conditions $u(0) = b$ and $u'(\pi) = c$.
 (a) Set up a finite difference scheme using centered difference approximations to solve the BVP using $n + 1$ grid points $t_i = ih$ with $i = 0, \dots, n$ and $h = \pi/n$. Put your system into matrix form.
 (b) Describe and set up another method of solving the BVP (not finite differences).

Problema 6 — Error del trapecio compuesto como integral de función de influencia

Buscamos G lineal a trozos tal que $I_n(g) - \int_0^1 g \, dx = \int_0^1 G(x)g'(x)dx$ para todo $g \in C^1[0, 1]$.

Derivación: en cada intervalo $[x_k, x_{k+1}]$ (con $x_{k+1} - x_k = \Delta_k$), aplicamos FTC a $\int_{x_k}^{x_{k+1}} (g(x)G(x))' dx = [g(x)G(x)]_{x_k}^{x_{k+1}}$:

$$\int_{x_k}^{x_{k+1}} G(x)g'(x)dx + \int_{x_k}^{x_{k+1}} G'(x)g(x)dx = [gG]_{x_k}^{x_{k+1}}.$$

Queremos que la contribución de $[x_k, x_{k+1}]$ a $I_n(g) - \int_0^1 g \, dx$ sea $\int_{x_k}^{x_{k+1}} G(x)g'(x)dx$. La contribución del trapecio compuesto es $\frac{\Delta_k}{2}(g(x_k) + g(x_{k+1}))$, y la integral exacta es $\int_{x_k}^{x_{k+1}} g \, dx$.

Si $G(x) = a_k x + b_k$ en $[x_k, x_{k+1}]$, $G'(x) = a_k$ (constante), entonces

$$\int_{x_k}^{x_{k+1}} G(x)g'(x)dx = [gG]_{x_k}^{x_{k+1}} - a_k \int_{x_k}^{x_{k+1}} g \, dx.$$

Igualando a $\frac{\Delta_k}{2}(g_k + g_{k+1}) - \int_{x_k}^{x_{k+1}} g \, dx$:

$$g(x_{k+1})G(x_{k+1}) - g(x_k)G(x_k) - a_k \int_{x_k}^{x_{k+1}} g \, dx = \frac{\Delta_k}{2}(g_k + g_{k+1}) - \int_{x_k}^{x_{k+1}} g \, dx.$$

Con $a_k = 1$: $g_{k+1}G(x_{k+1}) - g_k G(x_k) = \frac{\Delta_k}{2}(g_k + g_{k+1})$, que se satisface con $G(x_k) = -\Delta_k/2$ y $G(x_{k+1}) = \Delta_k/2$... Pero G debe ser continua y consistente en los nodos.

La solución correcta (verificable por la pista del enunciado): en cada $[x_k, x_{k+1}]$, $G(x) = x - \frac{x_k + x_{k+1}}{2}$ (desviación del punto medio del subintervalo). Verificación: $G(x_k) = x_k - \frac{x_k + x_{k+1}}{2} = -\frac{\Delta_k}{2}$, $G(x_{k+1}) = \frac{\Delta_k}{2}$.

$$\int_{x_k}^{x_{k+1}} G(x)g'(x)dx = [Gg]_{x_k}^{x_{k+1}} - \int_{x_k}^{x_{k+1}} G'g \, dx = \frac{\Delta_k}{2}g_{k+1} + \frac{\Delta_k}{2}g_k - \int_{x_k}^{x_{k+1}} g \, dx,$$

(usando $G' = 1$ y que $G(x_k) = -\Delta_k/2$, $G(x_{k+1}) = \Delta_k/2$).

Sumando sobre todos los subintervalos:

$$\int_0^1 G(x)g'(x)dx = \sum_k \left[\frac{\Delta_k}{2}(g_k + g_{k+1}) - \int_{x_k}^{x_{k+1}} g \, dx \right] = I_n(g) - \int_0^1 g \, dx. \blacksquare$$

$$\boxed{G(x) = x - \frac{x_k + x_{k+1}}{2} \text{ en } [x_k, x_{k+1}].}$$

ENUNCIADO — PROBLEMA 7

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Problema 7 — BVP con convección: diferencias finitas y método alternativo

$$-u'' + au' = (1 + \sin t)u, \quad t \in [0, \pi], \quad u(0) = b, \quad u'(\pi) = c.$$

(a) Diferencias finitas centradas. Nodos $t_i = ih$, $h = \pi/n$, $i = 0, \dots, n$. Condición Dirichlet en $t = 0$: $u_0 = b$ (conocido). Condición Neumann en $t = \pi$: $u'(\pi) \approx (u_n - u_{n-1})/h = c$ (diferencia hacia atrás), ó por nodo fantasma $(u_{n+1} - u_{n-1})/(2h) = c \Rightarrow u_{n+1} = u_{n-1} + 2hc$.

Para $i = 1, \dots, n-1$ (nodos interiores):

$$-\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} + a\frac{u_{i+1} - u_{i-1}}{2h} = (1 + \sin t_i)u_i.$$

Para $i = n$ (con $u_{n+1} = u_{n-1} + 2hc$):

$$-\frac{u_{n-1} - 2u_n + (u_{n-1} + 2hc)}{h^2} + a\frac{(u_{n-1} + 2hc) - u_{n-1}}{2h} = (1 + \sin t_n)u_n,$$

que simplifica a:

$$\frac{-2u_{n-1} + 2u_n}{h^2} - \frac{2c}{h} + ac = (1 + \sin t_n)u_n.$$

En forma matricial $A\mathbf{u} = \mathbf{f}$ con $\mathbf{u} = (u_1, \dots, u_n)^T$ ($u_0 = b$ conocido): La matriz A es tridiagonal de tamaño $n \times n$:

$$A_{i,i} = \frac{2}{h^2} + (1 + \sin t_i), \quad A_{i,i-1} = -\frac{1}{h^2} - \frac{a}{2h}, \quad A_{i,i+1} = -\frac{1}{h^2} + \frac{a}{2h},$$

con modificaciones en las filas correspondientes a los nodos de frontera.

(b) Método de shooting. Reescribir como sistema de primer orden: $y_1 = u$, $y_2 = u'$: $y'_1 = y_2$, $y'_2 = ay_2 - (1 + \sin t)y_1$, con $y_1(0) = b$, $y_2(0) = s$ desconocido.

Integrar el sistema IVP desde $t = 0$ hasta $t = \pi$ con un método numérico (Runge-Kutta) para obtener $y_2(\pi; s)$. Definir $F(s) = y_2(\pi; s) - c$ y resolver $F(s^*) = 0$ con bisección o Newton (con $F'(s) = \partial y_2(\pi; s)/\partial s$ obtenido integrando las ecuaciones de sensibilidad). Una vez encontrado $s^* = u'(0)$, la solución del IVP correspondiente da $u(t)$ en toda la malla.